Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Zheng, Chiang, and Sheng et al. (2023)

Hins Fang Charles Costanzo

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Tried before, but never studied systematically.



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That is, how well do LLM judges match **human preferences**?

- (and how do we even **measure** that?)

Meet the models: camelids in the hot seat



LLaMA-13B (Meta, 2023)



Vicuna-13B (LMSYS, 2023)

- These two are just our example. The paper evaluates up to 15 chatbots, including GPT-4 and Claude.

Photos: PA (llama) and Marshallhenrie (vicuña), Wikimedia Commons, CC BY-SA 4.0

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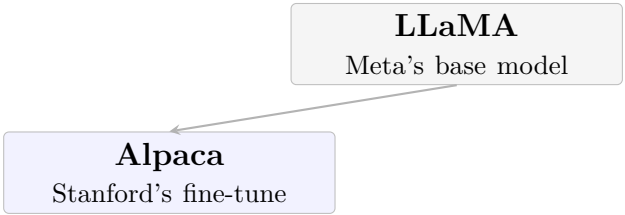
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LLaMA
Meta's base model

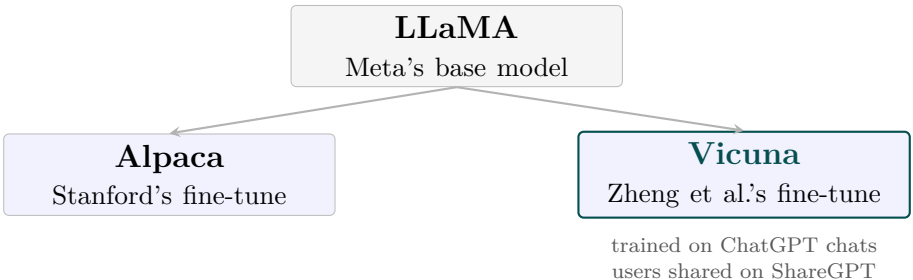
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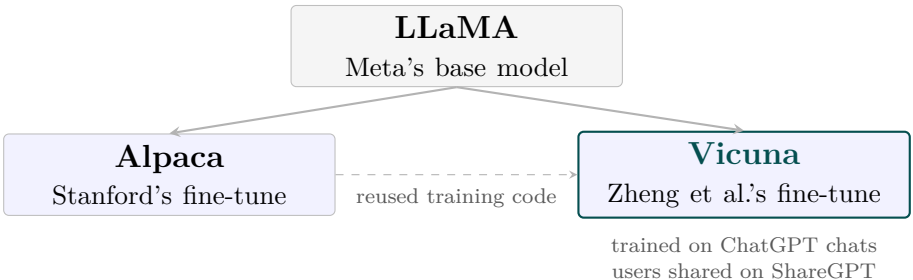
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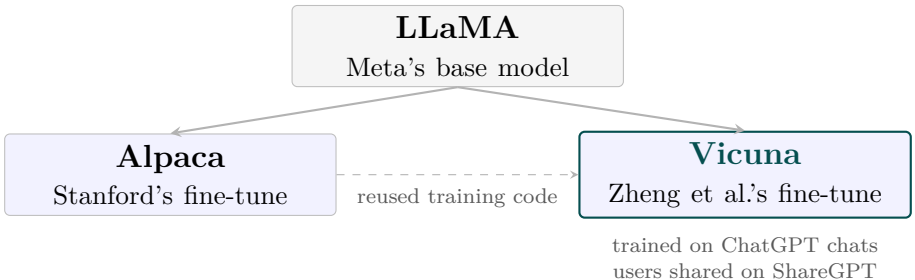
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Meet the models: Camelids in the Hot Seat

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Both are LLaMA's camelid descendants.

Meet the judges: LLMs on the bench

GPT-4



Meet the judges: LLMs on the bench



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Meet the judges: LLMs on the bench



- All three are checked against **human** judges:
  - 58 experts (MT-Bench) and crowdsourced ratings (Chatbot Arena)
- The judges are also among the models evaluated, so they sometimes grade their own answers.

Example 1:

Question

If the FED buys bonds in the secondary market

(A) the money supply will increase.
(B) the money supply will decrease.
(C) the money supply will not be affected.
(D) the discount rate would be affected.

Output the answer only.

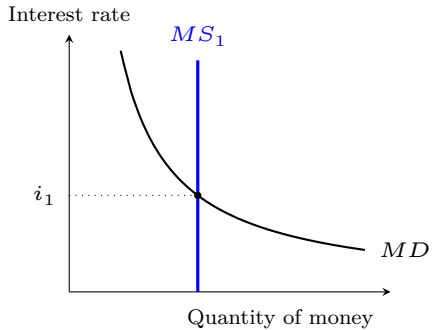
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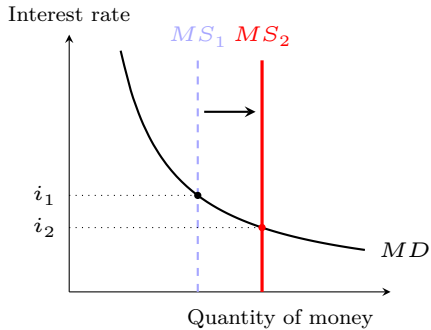
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Example 1

LLaMA-13B

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The Federal Reserve buys bonds in the secondary
market to increase the money supply.
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Vicuna-13B

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Example 1 - Follow-up:

Follow-up Question

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How does it affect my daily life? Give 3 examples
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How does it affect my daily life? Give 3 examples

LLaMA-13B

The Federal Reserve buys bonds in the secondary market to increase the money supply. This affects your daily life by:

1. Increasing the money supply.
2. Increasing the money supply and lowering the discount rate.
3. Increasing the money supply and lowering the discount rate and increasing the money supply.

Vicuna-13B

When the Federal Reserve buys bonds in the secondary market, it can have several effects on the economy that can impact your daily life. Here are three examples:

1. Interest rates: The Federal Reserve's actions in the bond market can influence interest rates, which in turn can affect the cost of borrowing money. For example, if the Federal Reserve buys bonds and drives up demand for them, interest rates on things like mortgages, car loans, and credit cards may go up, making it more expensive to borrow money.
2. Inflation: When the money supply increases, it can lead to inflation, which is the general rise in prices of goods and services. This can impact your daily life by making it more expensive to buy things like groceries, gas, and other necessities.
3. Employment: The Federal Reserve's actions in the bond market can also have an impact on employment. For example, if the Federal Reserve's actions lead to economic growth and increased demand for goods and services, it can create more job opportunities and boost employment.

You be the judge



menti.com code **1479 0094**

Which assistant did better in this conversation?

- LLaMA-13B
- Vicuna-13B
- Tie

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Then we'll see what GPT-4 said.

Example 1 - GPT-4 as judge

GPT-4 Judgment

LLaMA-13B provided an incorrect response to the user's question about how the Federal Reserve buying bonds in the secondary market affects daily life. The answer given is repetitive and lacks clear examples of how the action impacts daily life.

On the other hand, Vicuna-13B provided a relevant and accurate response to the user's question about the Federal Reserve buying bonds. The answer includes three clear examples of how the action impacts daily life, such as interest rates, inflation, and employment.

LLaMA-13B should improve by providing relevant, clear, and accurate examples to address the user's question. **Vicuna-13B's response is more helpful, relevant, and detailed.**

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Vicuna more helpful, relevant, and detailed

- LLaMA: repetitive, no real examples
- Vicuna: three clear, relevant examples

Look back at turn 1

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The prompt said to "output the answer only."

- LLaMA provided its own answer.
- Vicuna outputted the correct answer verbatim from the four options listed.

Scores under different benchmarks

Example 1 is an MMLU[1] question.

Model	MMLU (5-shot)	MT-Bench score (GPT-4)
LLaMA-13B	47.0	2.61
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Table 8 in the paper. MT-Bench is scored 1–10 by GPT-4.

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MMLU was not designed to measure human preferences.

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Key topics

- **Ground truth:** collecting human preferences to evaluate LLM judges against
  - **MT-Bench**: 80 multi-turn questions, 3,000 expert votes
  - **Chatbot Arena**: crowdsourced anonymous battles, 30,000 conversations

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- **Limitations:** LLM biases and reasoning errors
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- **Agreement:** how often LLM judges match human preferences

Two new benchmarks

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  - No single right answer to check against

Two new benchmarks

New primary evaluation metric:
human ratings

MT-Bench · Chatbot Arena

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- Judged by **58 experts**

MT-Bench: what a question looks like

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Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions.

Math, turn 1

Given that $f(x) = 4x^3 - 9x - 14$, find the value of $f(2)$.

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Writing, turn 2

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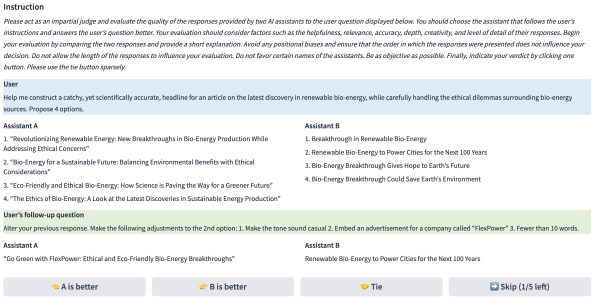
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Following a new instruction while building on its **own** previous answer.

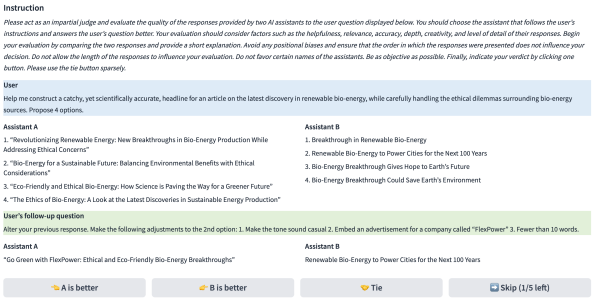
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How the experts voted



Voting interface, Figure 17 in Zheng et al. (2023).

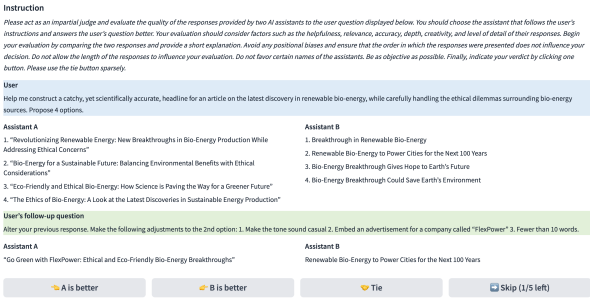
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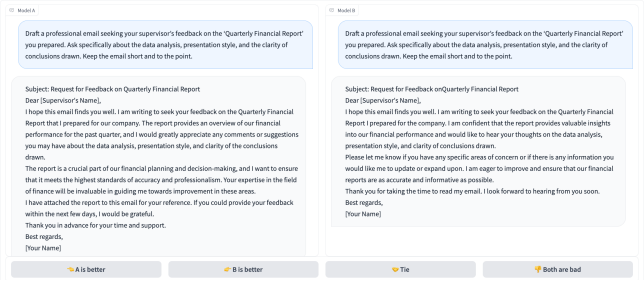
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- So humans and the LLM judge are graded on the **same rules**.

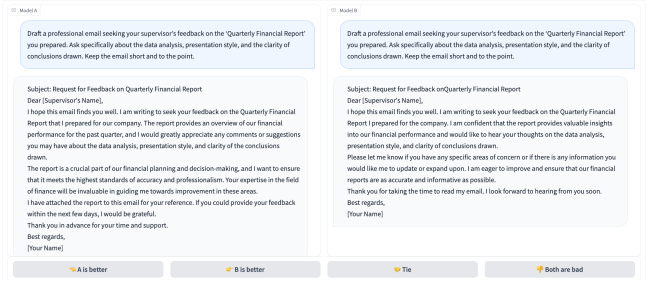
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Chatbot Arena: battles in the wild



Screenshot cropped from Figure 19 in Zheng et al. (2023).

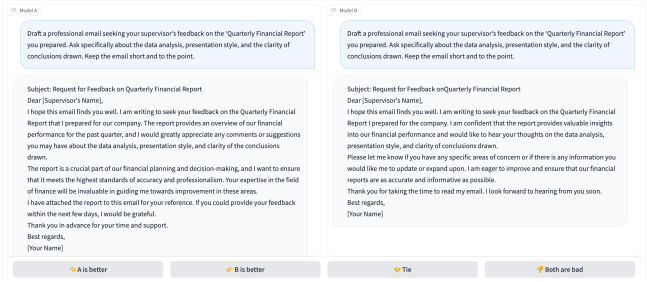
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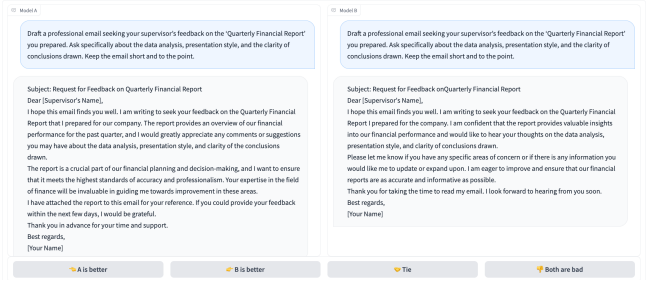
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- About 30,000 votes in the first month.
- Your poll was a mini Arena, except you knew which model was which. **Why hide the names?**

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Two kinds of ground truth

	MT-Bench	**Chatbot Arena**
Questions	Fixed, curated	Whatever users ask
Judges	58 experts	Crowd
Votes	About 3,000	About 30,000
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Now we have human verdicts. Can an LLM reproduce them, and where does it go wrong?

How an LLM judge is asked

- **Pairwise:** question + two answers $\rightarrow$ “A is better,” “B is better,” or tie

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Answers are labeled only "Assistant A" and "Assistant B."
The judge is **never told** which model wrote which answer.

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The default prompt already says to ignore position, length, and names.

Position bias

- Test: two near-identical answers to each of 80 questions, judged in both orders

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GPT-4	65.0%	30.0%

- Most judges favor the **first** answer. Even GPT-4 is inconsistent a third of the time.

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Verbosity bias

- “Repetitive list” attack on 23 answers that contain a list

Table 3, Zheng et al. (2023). Example in Figure 12.

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- Does the judge prefer the padded version?

	Claude-v1	GPT-3.5	GPT-4
Fooled by the padding	91.3%	91.3%	8.7%

Table 3, Zheng et al. (2023). Example in Figure 12.

Self-enhancement bias: what the paper found

- Compared each judge's win rates to the human win rates

Figure 3(b), non-tied first-turn votes, Zheng et al. (2023).

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The authors: "our study cannot determine whether the models exhibit a self-enhancement bias."

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- The paper's data cannot tell these apart, or rule out that the answers really were better

How strong is the evidence?

Test	Sample
Position bias	80 questions
Verbosity bias	23 answers
Math grading	10 questions
Self-enhancement	win rates for 6 models

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- **Your turn:** how would you test recognition vs. style preference?

Limited reasoning: can you grade this?

Question

Benjamin bought 5 sci-fi novels at $20 each, 3 history books at $30 each, and 2 philosophy books at $45 each. What was the total cost?

Assistant A

The total cost of Benjamin's purchases was $115.

Assistant B

The total cost of his purchases is $295.

Question shortened from Figure 13, Zheng et al. (2023)

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GPT-4 judgment

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Asked the question on its own, GPT-4 answers $280. It was **misled by the answers**.

Question shortened from Figure 13, Zheng et al. (2023)

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  - But prompts cost 4× more, and the examples may add new biases

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- **Chain of thought:** solve the problem first, then grade

Table 4: 10 questions, each judged in both orders, Zheng et al. (2023).

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Verbosity and self-enhancement: no fix proposed.

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References

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